Project Name: [OnCanteen]

Team Members:

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Problem (The "Why")

Employees working on the upper floors of Formulatrix face significant hurdles when trying to buy lunch at the ground-floor canteen. The physical distance from stairs and elevators, combined with long queues during peak hours, heavily consumes their limited break time. Furthermore, because the canteen operates on a buffet system, employees cannot see the daily menu without physically going downstairs. Consequently, they often rely on external food delivery apps, causing the canteen to lose potential customers. The OnCanteen app bridges this gap by digitizing menu visibility and the ordering process. For Formulatrix, this boosts employee productivity by eliminating wasted break time. Additionally, the pre-order system minimizes the risk of employees exceeding their allotted lunch hour.

- **Current Issue:** Limited lunch break duration consumed by walking down stairs and waiting in long canteen queues.
- **Impact:**
    1. **UMKM Canteen:** Increases daily order volume, reduces lost sales, and supports business growth.
    2. **For Employees:** Eliminates wait times, saves money on delivery fees, and maximizes their actual resting time.

Proposed Solution

- **Our Approach:** We will develop **OnCanteen** as a Progressive Web App (PWA) using React PWA as mobile Front End and C# Web API as Backend. This approach delivers a native mobile app experience that is lightweight, fast, and accessible directly via a web browser without the friction of downloading from an app store.

- **Key Value:** Providing real-time menu visibility, seamless digital pre-ordering, and a "skip-the-queue" experience to save employees' break time while maximizing canteen revenue.

Core Features

- **Feature 1:** List Menu Food
- **Feature 2:** Payment Gateway
- **Feature 3:** Cart Foods
- **Feature 4:** Add Food and Update stock
- **Feature 5 :** Register User
- **Feature 6 :** Login User

High-Level Architecture & Quality Assurance

Decoupled system using React (PWA) for a highly responsive, app-like frontend experience, integrated via RESTful endpoints to a .NET Web API backend.

- **Architecture Pattern:** Clean Architecture
- **Design Patterns Used:** Repository Pattern and Dependency Injection
- **Code Quality Goal:** OOP concepts and SOLID Principles.
- **Quality Assurance (QA) :** Implement unit testing xUnit and Moq

Identified Challenges & Risks

- **Risk 1:** Learning Curve on PWA Implementation. As this is the first time building a Progressive Web App using React, there is a risk of technical delays, especially given the tight 2-week development timeline.
- **Risk 2:** Payment Gateway Production Readiness. Midtrans requires a verified legal entity for production deployment. We will mitigate this by using Midtrans Sandbox for the MVP phase.
- **Risk 3:** Real-Time Inventory Discrepancy. Ensuring the digital stock in the app perfectly matches the physical food available in real-time, preventing users from ordering sold-out items.
- **Risk 4 :** Fulfillment Priority Conflict. Establishing a clear Service Level Agreement (SLA) for the canteen staff on how to prioritize digital pre-orders versus the physical walk-in queue during peak lunch hours.
- **Risk 5 :** High-traffic concurrency during lunch hour

- **Risk 6 :** App update delays by canteen staff